

MARK MANOTI NDEGE

Software Engineer | Full-Stack Developer | Systems & Technical Support

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SUMMARY

Computer Science diploma student and Software Engineering Intern at Spin Mobile, with two years of hands-on, production-grade full-stack development as Co-Founder & CTO of [SmartShamba](#), an agri-tech platform live across 9+ Kenyan counties. Comfortable across the stack — Python, Java, TypeScript/Next.js, FastAPI — with applied experience in AI/ML, cybersecurity tooling, and USSD/mobile-money integrations for real users. Strong grounding in OS architecture, networking fundamentals, and systems design supports technical support and IT operations roles alongside development.

Seeking a role in software engineering, systems development, or technical support — including IT/network operations and technical troubleshooting.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, SQL.

Frameworks & Libraries: Next.js, React, FastAPI, Django, Tailwind CSS, Prisma ORM.

Databases & Infra: PostgreSQL (Supabase, pgvector), SQLite, Redis, Docker, Celery.

AI/ML & Data: Scikit-learn, RAG & vector search, Jupyter, Google GenAI SDK, Claude API.

Networking & Security: Cisco networking fundamentals (certified), TCP/IP, threat/malware detection.

OS & Hardware: Computer Organization & Architecture, Operating Systems fundamentals, Linux (Ubuntu), hardware troubleshooting.

Tools & Platforms: Git/GitHub, VS Code, PyCharm, Cisco Packet Tracer, Linux (Ubuntu).

Algorithms & Systems: State machines, matching algorithms, event-driven architecture, DAO design pattern, Big-O analysis.

Graphic Design & UI/UX: Responsive UI layout, Tailwind CSS design systems, Figma basics, web typography, HTML5/CSS3

Integrations: M-Pesa Daraja API, Africa's Talking USSD/SMS.

EXPERIENCE

Software Engineering Intern — Spin Mobile LLC, Nairobi

Aug 2026 – Present

- Selected for a 12-week full-stack engineering internship covering Python, Django, and PostgreSQL under senior engineer mentorship
- Deliver weekly graded assignments (CLI tooling, data validation, automated testing) evaluated on code quality, daily discipline, and a capstone defense
- Maintained 100% build pass rates and clean code standards across weekly deployments using branch-protected Git workflows, and PyCharm-based development

Co-Founder & CTO — SmartShamba, Nairobi

April 2025 – Present

[smartshamba.vercel.app](#)

- Co-founded and lead technical development of a production agri-tech platform digitizing maize trading, now serving 9+ counties in Kenya.
- Hybrid USSD/Web Transaction System: enables low-connectivity farmers to discover buyers, list produce, and record transactions without a smartphone, synchronized on a single Supabase PostgreSQL ledger.
- Transaction State Machine: controls the full lifecycle from offer and negotiation through confirmation, delivery, settlement, dispute resolution, and completion.
- Architected a unified transaction ledger on Supabase PostgreSQL synchronizing feature-phone farmers (via Africa's Talking USSD) with institutional buyers (via web), with M-Pesa Daraja payment integration.
- Built and shipped the platform end-to-end on Next.js 16, TypeScript, and Prisma ORM, including dispute-resolution/ratings and group-selling feature sets.
- Own the full engineering lifecycle — architecture, API design, USSD flows, admin/farmer UI, and error monitoring with Sentry.

FEATURED PROJECTS

AETSH-69 — FastAPI & React AI Platform — Personal

aetsh69.vercel.app

- Built a full-stack RAG platform with FastAPI, PostgreSQL + pgvector semantic search, Redis caching, and Celery workers, integrating the Claude API for sub-second semantic search
- JWT-authenticated, multi-tenant architecture with a React/TypeScript frontend

DEEP-TRIO — ML Malware Detection Scanner — Personal

- Trained and deployed Scikit-learn classifier models to detect malicious payloads, with an automated quarantine/isolation pipeline and a web interface

Spendy — CLI Expense Tracker — Personal / Mentorship project

github.com/N3stah/spendy

- Built a modular Python CLI tool with a functional architecture separating logic from I/O; reached 100% test coverage on pure functions with Pytest

Java CLI Task Manager — Personal

github.com/N3stah/task-manager

- Enterprise-structured Java CLI application using the DAO design pattern, SQLite via JDBC, and Maven Shade packaging

Full project suite, including a 22-project Python engineering portfolio, at aetsh69.duckdns.org/portfolio

EDUCATION

The Nairobi National Polytechnic

May 2024 – May 2027 (expected)

Diploma in Computer Science (Level 6)

CBET programme: Algorithms & Data Structures, Operating Systems, Computer Organization & Architecture, Database Management Skills, Networking & Distributed Systems, Information Systems, Fundamentals of Programming, Web & Graphic Design, Mathematics for CS.

Pope Paul VI Junior Seminary

2020 – 2023

Kenya Certificate of Secondary Education (KCSE)

CERTIFICATIONS

- Intermediate AI & Machine Learning — University of Nairobi, KITI (Apr 2026)
- Foundation AI & Machine Learning — University of Nairobi, KITI (Apr 2026)
- Cybersecurity Foundation — University of Nairobi, KITI (Apr 2026)
- Blockchain Foundation — University of Nairobi, KITI (Apr 2026)
- Internet of Things (IoT) Foundation — University of Nairobi, KITI (Apr 2026)
- KITI Program Bootcamp II — Emerging Tech — University of Nairobi C4D Lab (Apr 2026)
- KITI Project Hackathon II — University of Nairobi C4D Lab (Apr 2026)
- Getting Started with Cisco Packet Tracer — Cisco Networking Academy (Mar 2026)

LEADERSHIP & ACTIVITIES

- Secretary, Young Christian Students (YCS)
- Represented school at the Kenya Music and Drama Festival

REFEREES

• Mr. Fred Obadha

Lecturer, Department of ICT (Information and Communication Technology)

The Nairobi National Polytechnic.

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